PORA TECH ACADEMY

Data Analytics Course

Revenue and Growth Capstone Project Report

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Project Overview

The analysis was carried out to evaluate the revenue for Olist Brazilian E-Commerce for the years 2016 to 2018, focusing on five KPIs: Monthly GMV trend, Revenue by state, Payment method breakdown, Year-over-year GMV comparison, and credit-card installment analysis. We also added interactive filters for Year, State, and Order Status to see how these factors change the revenue.

Data used and Overview

For the analysis, we loaded four (4) of the Olist tables into the SQLite database to have a full view of who bought, what was ordered, how it was paid for, and when each order happened. This is the data that powers the Revenue & Growth Streamlit dashboard.

Olist Dataset	Contents	Uses
Customers	Customer ID and location (city, state)	Geographic analysis
Order items	Items, price, and freight per order	Product revenue
Order Payment	Payment type, instalments, value	Total revenue and payment mix
Orders	Order status and timestamps	Order volume and trends

Cleaning Methodology

What changed

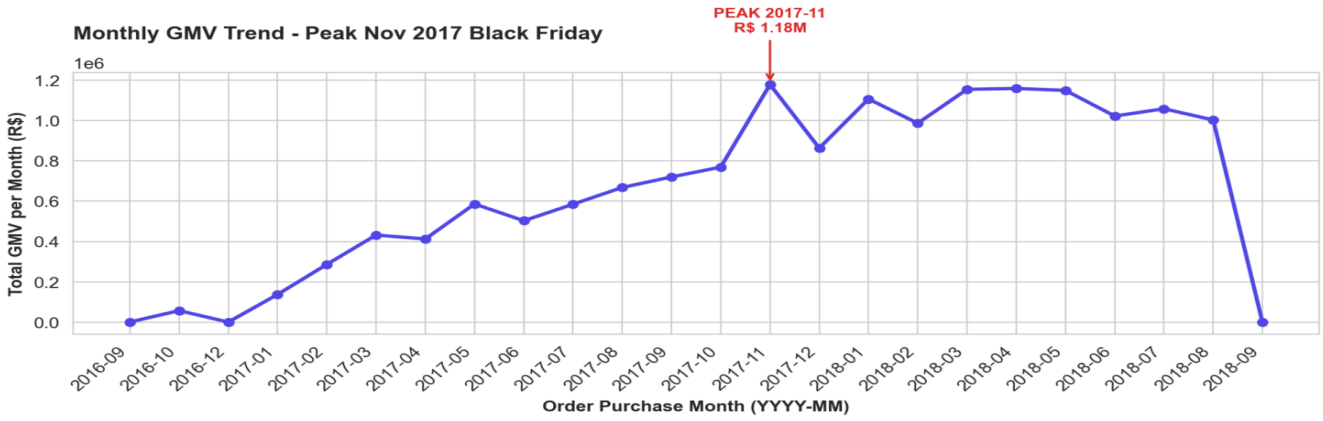
- **Column name:** Using the strip() function, leading and trailing whitespace is removed before writing each table to the database.
- The dataframe was exported to the database using “df.to_sql(table_name, conn, if_exists="replace", index=False).”
- **Instalment type**: We cast the payment installment, initially stored as text, to an integer in the query.

What we added

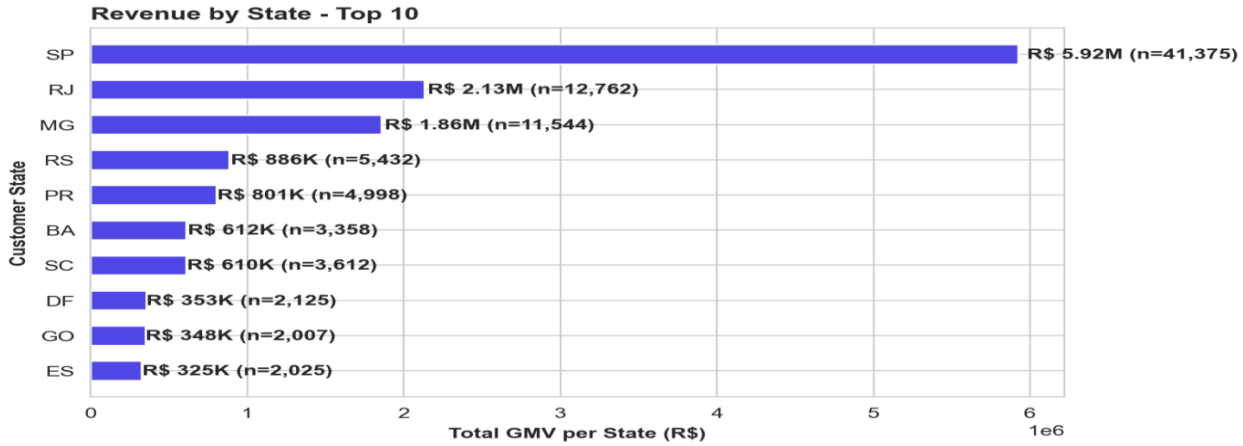
- **Order status filter:** Our base GMV calculation includes every order's price plus freight, regardless of status. We kept the raw data untouched but added a filter in the dashboard so users can isolate specific statuses (like 'delivered') if they want.

Results and Insight

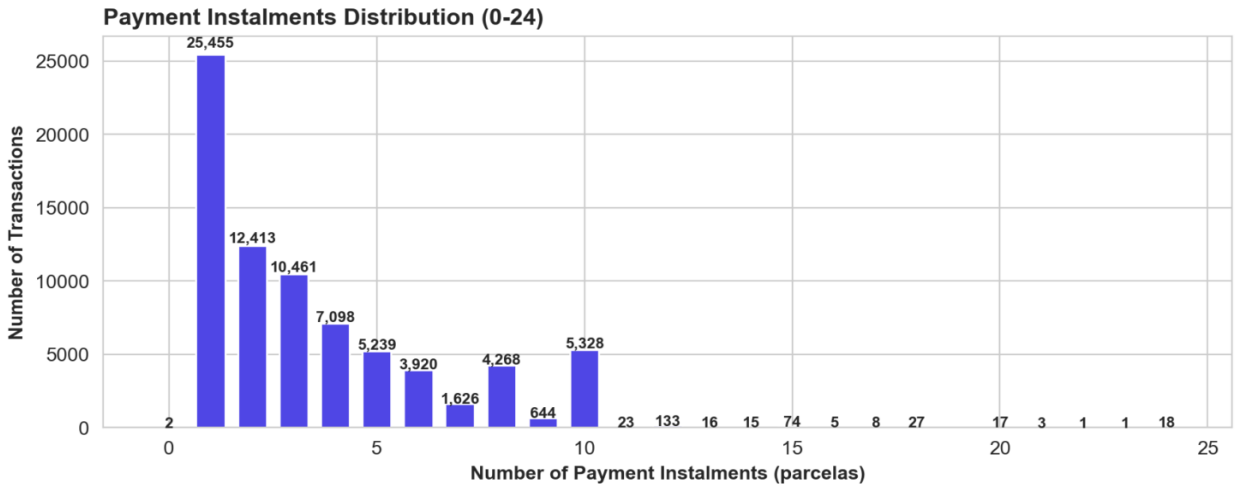
1. **Insight:** Our Monthly GMV Trend shows low sales in late 2016, steady growth through 2017, and a sharp peak in 2017-11 at R$1.18M during Black Friday. Total GMV is R$15.84M, with an Average Payment Value of R$154.1 and Credit Card Payment Type at 73.9%. When we filter by All states, SP state alone is 37.38% of Total GMV (R$5.92M) — showing high state concentration. (Note: The sharp drop at the end of the chart reflects the 2018 dataset ending in October)



2. **Insight:** Revenue is heavily concentrated in a few states. SP contributes 37.38% of Total GMV (R$5.92M), with RJ and MG far behind. This concentration means a problem in São Paulo logistics or demand would affect more than a third of the company's revenue.



3. **Insight:** Credit Card is the dominant payment type at 73.9% of transactions. Looking at the instalments, the vast majority of customers choose just 1 to 3 instalments. However, there is a long tail of customers splitting larger purchases across up to 24 months.



Limitations and Recommendations

1. **Incomplete Year Coverage:** 2016 has only Sep, Oct, and Dec (no November) - 3 months. 2018 stops at Oct 17. So, yearly GMV comparison is not balanced. We need full 12-month data for a fair YoY comparison.
2. **Cancelled Orders in GMV:** The default Total GMV (R$15.8M) includes 625 cancelled orders, making it higher than the actual Delivered GMV of R$15.4M. To solve this, we added an Order Status filter to the dashboard so users can easily isolate 'delivered' orders to see true revenue.
3. **No Customer Demographics:** We don't have age or income data, which means we cannot analyze customer lifetime value or retention beyond using payment type and state only.
4. **State Concentration Risk:** 37.38% of Total GMV is from SP state alone. If SP logistics fails, one-third of revenue is at risk. There's a need for growth in other states.
5. **Freight Included in GMV:** Total GMV includes freight_value. Freight goes to the logistics partner, not full margin for Olist. It is better to separate Item Revenue (R$13.5M) and Freight Revenue when reporting.